gpeyt1-06 · OC clay settlement

GEO PE · Long Calculation Problems I — Topic 2: Soil Mechanics, Lab Testing & Analysis

Problem

A warehouse slab will add a uniform Δσ = 1,200 psf (wide load) at the top of a 16 ft clay layer. Profile: 0–10 ft sand, γ = 115 pcf (water table at 10 ft, top of clay); 10–26 ft clay, γsat = 108 pcf, e0 = 1.05, Cc = 0.42, Cr = 0.06, σ′p = 1,450 psf; below is dense gravel. The clay's cv = 0.35 ft²/day and LL = 61 (not needed here). Use one layer evaluated at mid-depth.

Most nearly, the primary consolidation settlement of the clay is:

A. 10.6 in.
B. 10.0 in.
C. 1.4 in.
D. 8.0 in.

Given equations / data

> γw = 62.4 pcf; OC clay with σ′0 < σ′p < σ′f: S = [Cr H/(1+e0)] log(σ′p/σ′0) + [Cc H/(1+e0)] log(σ′f/σ′p)

Worked solution

1. σ′0 at mid-clay = 10(115) + 8(108 − 62.4) = 1,515 psf
2. σ′f = 1,515 + 1,200 = 2,715 psf > σ′p = 1,450 → crosses σ′p
3. Recompression: 0.06×16/2.05×log(1450/1515) = -0.0089 ft
4. Virgin: 0.42×16/2.05×log(2715/1450) = 0.8928 ft
5. S = 0.8840 ft = 10.6 in.

Final answer

10.6 in.

Why the other values are wrong

- **10.0 in.** — treated the clay as normally consolidated (Cc from σ′0)
- **1.4 in.** — used Cr for the whole stress range
- **8.0 in.** — used total stress at mid-clay

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.2.1–3.2.2 Consolidation (NC and OC soils), pp. 82–84

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=IGS94AWmPWA, https://www.youtube.com/watch?v=NwGztxby-vQ (no creator values copied).

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